

A Research Principle for the AI Era

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Research in the AI era should pursue intrinsic value while building the external support needed to sustain that pursuit.

Even when intrinsic value is the sole objective, an initial phase devoted entirely to external return can be necessary for optimal research. We prove this for a linear benchmark with a sufficiently long horizon, then illustrate the same pattern in one numerical example with nonlinear functions that change in every cycle.

Here, intrinsic value means the scientific understanding or other substantive contribution that the researcher ultimately values. External return means support convertible into research funding, measured in the same units as expenditure. The distinction separates the purpose of research from the resources that sustain it.

Assumptions

Consider a fixed number n of research cycles. In cycle i , let c_i be expenditure and $p_i \in [0, 1]$ the share allocated to intrinsic work. Write I_i and E_i for intrinsic value and external return, respectively.

1. Increasing production and funding functions between linear bounds. Choose units so that $c_1 = 1$, and write

$$I_i = p_i f_i(c_i), \quad E_i = (1 - p_i) g_i(c_i), \quad c_{i+1} = h_i(E_i).$$

Each function is strictly increasing. For each family $F = f, g, h$, there are constants $0 < \ell_F < L_F$ such that

$$\ell_F x \leq F_i(x) \leq L_F x \quad (x \geq 0, 1 \leq i \leq n).$$

The bounds apply to function values; the functions may be nonlinear and may change across cycles. They bound output per unit of input without fixing curvature. The proportion enters directly as p_i or $1 - p_i$. For the final cycle, the funding update is only a convenient notation.

2. A positive and growing safety balance. Let B_i denote the cumulative funding surplus at the end of cycle i :

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

Require

$$B_i \geq \max\{\delta, \rho B_{i-1}\}, \quad \delta > 0, \quad \rho > 1.$$

The fixed floor δ establishes a safety reserve, while ρ requires it to grow. Funding is assessed at cycle boundaries; the initial expenditure is supplied as seed funding.

3. Intrinsic value as the sole objective. Choose the allocation shares to maximize

$$\sum_{i=1}^n I_i.$$

All functions are deterministic and known when the allocation is chosen. Each cycle's intrinsic value receives equal weight, and external return has value through the funding dynamics and constraints.

An exact result for the linear benchmark

For the theorem, specialize to

$$f_i(c) = c, \quad g_i(c) = bc, \quad h_i(e) = e,$$

with $0 < \delta \leq b - 1$ and $1 < \rho < b$. These inequalities make the initial reserve feasible and leave room for intrinsic work while the reserve grows. Define

$$K = \min \left\{ m \geq 1 : (b - \rho) \sum_{j=0}^{m-1} \rho^j > 1 \right\}, \quad n > K.$$

Here m is an integer. Since $b > \rho > 1$, K is finite. The horizon condition gives early external returns enough time to generate later intrinsic value.

Theorem 1. *In this linear benchmark, every optimal strategy devotes the first $n - K$ cycles entirely to external return:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

There is an optimal strategy that then meets the balance-growth constraint exactly in all remaining cycles:

$$B_i = \rho B_{i-1}, \quad p_i = 1 - \frac{1 + \rho B_{i-1}}{b(1 + B_{i-1})} \in (0, 1) \quad (n - K < i \leq n).$$

Thus, maximizing intrinsic value alone requires an initial phase of full external focus and admits a single subsequent switch to a mixture of intrinsic and external work.

A numerical example

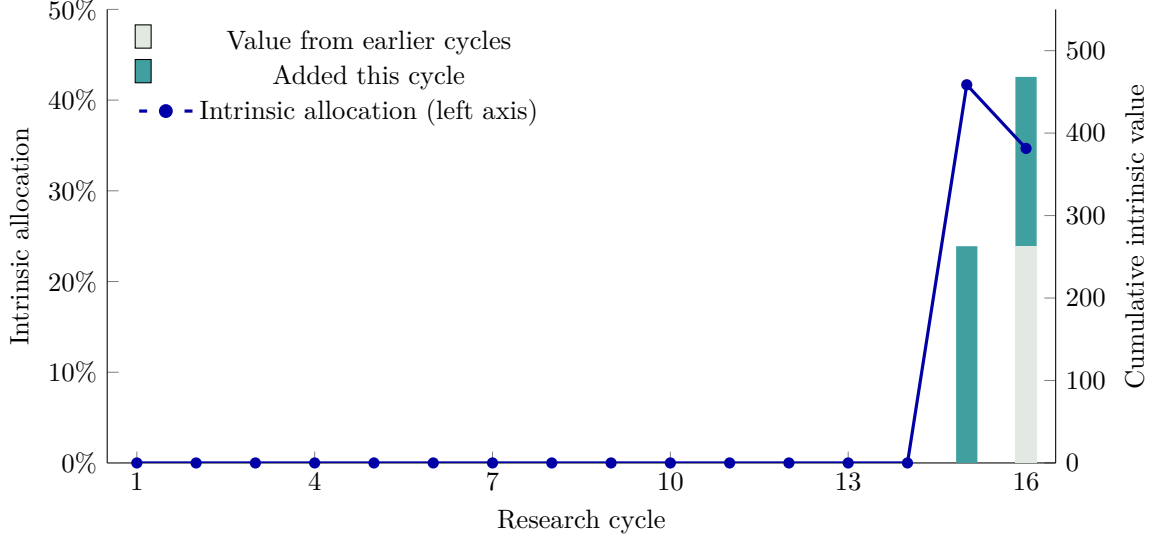


Figure 1: One numerically optimized trajectory with three nonlinear functions changing in every cycle. The line shows the intrinsic allocation share (left axis). Each full bar shows cumulative intrinsic value (right axis); its darker segment is the intrinsic value added in that cycle, and its lighter segment is the value accumulated in earlier cycles.

We use one fixed sequence of nonlinear functions with $n = 16$, $\rho = 1.05$, and $\delta = 0.1$. Every f_i , g_i , and h_i has the form

$$F_i(x) = a_i x \left(1 + q_i \frac{x}{1+x} \right), \quad 0.05 \leq q_i \leq 0.15.$$

The coefficients a_i lie in $[0.9, 1.1]$ for f_i , $[1.25, 1.65]$ for g_i , and $[0.85, 0.95]$ for h_i . Both coefficients vary across cycles and across the three function families. For every $x \geq 0$,

$$a_i x \leq F_i(x) \leq a_i(1 + q_i)x, \quad F'_i(x) = a_i \left[1 + q_i \left(1 - \frac{1}{(1+x)^2} \right) \right] > 0.$$

Thus these are strictly increasing nonlinear functions between strictly increasing linear bounds. Uniform lower and upper slopes are 0.9 and 1.265 for f_i , 1.25 and 1.8975 for g_i , and 0.85 and 1.0925 for h_i .

The coefficients are generated once with seed 20260928 and then held fixed and known throughout the optimization. A constrained numerical solver yields full external focus for cycles 1–14. The intrinsic allocation shares in cycles 15 and 16 are 41.7% and 34.7%; the corresponding intrinsic increments are 262.84 and 205.35, giving a final total of 468.19 in normalized intrinsic-value units. The safety floor and balance-growth constraints hold throughout this trajectory.

This is one numerical illustration of the nonlinear model. The exact theorem above concerns the linear benchmark; the numerical solution does not establish a general theorem or certify global optimality for the nonlinear model.

Proof

For the linear benchmark, the funding identity telescopes to

$$B_i = c_{i+1} - 1, \quad c_i = 1 + B_{i-1}.$$

Consequently, at any cycle after the first, a current balance $B \geq \delta$ permits exactly the next balances B' satisfying

$$\rho B \leq B' \leq bB + b - 1.$$

The intrinsic value earned in that cycle is

$$I = 1 - \frac{1}{b} + B - \frac{B'}{b}.$$

The upper endpoint corresponds to $p = 0$; the lower endpoint meets the required balance growth exactly.

Let $V_m(B)$ be the largest total intrinsic value attainable with m cycles remaining and current balance $B \geq \delta$. We show by induction that V_m is affine in B , and denote its slope by A_m . Since $V_0 = 0$, $A_0 = 0$. The recursion is

$$V_m(B) = 1 - \frac{1}{b} + B + \max_{\rho B \leq B' \leq bB + b - 1} \left\{ V_{m-1}(B') - \frac{B'}{b} \right\}.$$

If V_{m-1} is affine, the coefficient of B' is $A_{m-1} - 1/b$. Therefore the maximizing choice is the upper endpoint when $A_{m-1} > 1/b$, and the lower endpoint when $A_{m-1} < 1/b$. At equality, either endpoint is optimal; choose the lower one. Substitution shows that V_m is affine and gives

$$A_m = \begin{cases} 1 - \rho/b + \rho A_{m-1}, & A_{m-1} \leq 1/b, \\ b A_{m-1}, & A_{m-1} > 1/b. \end{cases}$$

Until the threshold is crossed,

$$A_m = \frac{b - \rho}{b} \sum_{j=0}^{m-1} \rho^j.$$

Hence $A_{K-1} \leq 1/b < A_K$. After this crossing, the second branch keeps $A_m > 1/b$. It follows that every cycle with more than K cycles remaining has the unique optimal choice of full external focus. With at most K cycles remaining, choosing the lower endpoint is optimal.

For the first cycle, the feasible interval is $\delta \leq B_1 \leq b - 1$. Its objective has coefficient $A_{n-1} - 1/b$ on B_1 . Since $n > K$, this coefficient is strictly positive, so every optimum chooses $B_1 = b - 1$, equivalently $p_1 = 0$. The same argument applies to the subsequent cycles with more than K cycles remaining.

In the last K cycles, choose $B_i = \rho B_{i-1}$. The formula for p_i follows from $E_i = 1 + B_i = b(1 - p_i)(1 + B_{i-1})$. Since $B_{i-1} > 0$ and $1 < \rho < b$, this share lies strictly between zero and one. This proves the result. $\square$

Build external support early so that the full research trajectory can deliver more intrinsic value.